

Calculus Homework Assignment

Instructions

Show all steps clearly. Simplify your answers where possible.

Question 1: Limits

Evaluate the following limit:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

Question 2: Derivatives

Find the derivative of the function:

$$f(x) = 3x^3 - 5x^2 + 2x - 7$$

Question 3: Product Rule

Differentiate the function:

$$y = x^2 \cdot \sin(x)$$

Question 4: Integrals

Compute the indefinite integral:

$$\int (4x^3 - 2x + 1) dx$$

Question 5: Definite Integral

Evaluate the definite integral:

$$\int_0^2 (x^2 + 1) dx$$